

CHAPTER 13

PRINCIPLES OF TEST SELECTION AND ADMINISTRATION

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After completing this chapter, you will be able to

- identify and explain reasons for performing tests with athletes;
- understand testing terminology to communicate clearly with athletes and colleagues;
- evaluate the validity and reliability of a test and its metrics;
- select appropriate tests and metrics; and
- administer test protocols properly and safely.

The strength and conditioning professional with a broad understanding of exercise and sport science can effectively use tests and measurements to make training decisions that help athletes achieve their goals, maximize their performance potential, reduce injury risk, or facilitate recovery from fatigue and injury. Tests and measurements form the objective core of the athlete evaluation process. This chapter covers the reasons for testing, testing

terminology, evaluation of test quality, the selection of appropriate tests and metrics, and aspects of proper test administration.

Reasons for Testing

Testing helps athletes and coaches objectively assess athletic talent and identify physical abilities and areas in need of improvement. In addition, test scores can be used in goal setting by setting attainable benchmarks for athletes to aim for. Baseline measurements can be used to establish starting points against which achievable training goals can be set, and testing at regular intervals can help track an athlete's progress in reaching those goals and provide insight into fatigue, readiness, and injury risk. Using tests as a basis for goal setting allows coaches to set specific goals for individual athletes that, when taken together, help to accomplish group or team objectives (see [chapter 9](#) for more information about goal setting).

Assessment of Athletic Talent

It is important for a coach to determine whether an individual has the physical potential to play a sport at the competitive level of the team. That judgment is not difficult if the candidate has already excelled at the sport elsewhere and is of adequate body size. However, in many cases, candidates have not clearly demonstrated their competitive abilities or may lack experience in the sport. The coach then needs some way of determining whether the candidate has the necessary basic physical abilities that, in combination with technique training and practice, could produce a competitive player. Field tests serve as tools for such assessment, and this application of testing is referred to as **talent identification**.

Identification of Physical Abilities in Need of Improvement

While some physical abilities are innate and not amenable to change, other physical abilities can be improved through training. By using appropriate testing methods and analysis, the strength and conditioning professional can determine which physical qualities can be targeted by participation in prescribed exercise programs (28,

29). This can only be achieved with an understanding of normative data for key test metrics (see [chapter 14](#) for normative data examples); this allows the strength and conditioning practitioner to objectively compare athletes' performances to representative normative values and identify relative strengths and areas for improvement.

Evidence-Based Testing

Evidence-based testing is essential, because it ensures that training programs are based on scientific research and empirical evidence, which are both equally important. Some of the key reasons for this follow:

- **Validity and reliability.** Evidence-based testing provides a framework for selecting tests that are both valid and reliable. By using validated and reliable tests, practitioners can confidently assess an athlete's capabilities and track progress.
- **Individualization.** Every athlete has distinct strengths, weaknesses, and goals. Evidence-based testing allows practitioners to modify training programs to the specific needs of each athlete. By conducting assessments based on scientific evidence, strength and conditioning professionals can identify areas in need of improvement and develop targeted interventions to optimize performance.
- **Injury prevention.** Evidence-based testing helps identify biomechanically inefficient movement patterns and muscular imbalances that may predispose athletes to injury. By addressing these issues through targeted training interventions, strength and conditioning professionals can reduce the risk of injury and enhance athletic longevity.
- **Performance optimization.** Evidence-based testing enables strength and conditioning professionals to identify performance-limiting factors and develop strategies to optimize athletic performance. By assessing key physiological parameters such as strength, power, agility, and aerobic endurance, strength and conditioning professionals can design training programs that target specific areas for improvement, leading to enhanced athletic performance.
- **Monitoring progress.** Regular assessment is essential for monitoring progress and evaluating the effectiveness of training interventions. Evidence-based testing provides objective measures of performance that allows strength and conditioning professionals to track changes over time and make data-driven decisions regarding program modifications. This ensures that training programs remain aligned with performance goals.

- **Professional credibility.** In an increasingly competitive field, evidence-based practice enhances the credibility and reputation of strength and conditioning professionals. By basing their work in scientific evidence, practitioners demonstrate a commitment to excellence and professionalism.

A variety of reputable sources, such as the following, can be used to ensure that assessments and practices are grounded in scientific evidence:

- Peer-reviewed journals
- Academic textbooks
- Professional organizations
- Meta-analyses and systematic reviews
- Conferences and workshops
- Evidence-based practice databases
- In-house data collection

Screening for Fatigue and Injury Risk

Monitoring acute and chronic fatigue due to training and competition exposures and associated injury risk factors can allow the strength and conditioning professional to prospectively modulate training programs to promote athlete health and performance. Although both fatigue and injury risk mechanisms are complex and multifaceted, combining quantitative and qualitative tests can provide valuable insights for the strength and conditioning professional and inform any requirements for acute adjustments to predetermined training programs, particularly during the competition period.

Testing can be used to assess athletic talent, identify physical abilities and areas in need of improvement, set goals, and evaluate progress.

Testing Terminology

To communicate clearly with athletes and colleagues, strength and conditioning professionals should use consistent terminology. The following terms and definitions are widely accepted and are used in this text:

test—A procedure for assessing ability in a particular endeavor.

monitoring—Taking repeat measures of an athlete to help guide the training process with appropriate data.

field test—A test used to assess ability that is performed away from the laboratory and does not require extensive training or expensive equipment (11).

measurement—The process of collecting test data (15).

evaluation—The process of analyzing test results for the purpose of making decisions. For example, a strength and conditioning professional examines the results of physical performance tests to determine whether the athlete's training program is effective in helping achieve the training goals or whether modifications in the program are needed.

pretest—A test administered before the beginning of training to determine the athlete's initial basic ability levels. A pretest allows the strength and conditioning professional to design the training program in keeping with the athlete's initial training level and the overall program objectives.

midtest—A test administered one or more times during the training period to assess progress and modify the program as needed to maximize benefit.

formative evaluation—Periodic reevaluation based on midtests administered during the training, usually at regular intervals (2). It enables monitoring of the athlete's progress and adjustment of the training program according to the athlete's individual needs. It also allows evaluation of different training methods and collection

of normative data. Regular modification of the training program based on formative evaluation keeps the training program fresh and interesting and helps avoid physical and mental staleness.

posttest—Test administered after the training period to determine the success of the training program in achieving the training objectives.

Evaluation of Test Quality

Test results are valuable only when two key criteria are met: validity and reliability. Validity ensures the test accurately measures its intended attribute, while reliability guarantees consistency in measurement across multiple instances. These fundamental characteristics are essential in assessing the quality of a test and its associated metrics. Both must meet acceptable standards for the test to provide meaningful benefits.

Validity

Validity refers to the degree to which a test or test item measures what it is supposed to measure and is one of the most important characteristics of testing (2, 25). For tests of physical properties such as height and weight, validity is easy to establish. For example, close correspondence between the readings on a spring scale and the readings on a calibrated balance scale indicates validity of weighing with the spring scale. The validity of tests of basic athletic abilities or capacities is more difficult to establish. There are several types of validity, including construct validity, face validity, content validity, and criterion validity.

To be valid, physical performance tests should measure abilities important in the sport, produce repeatable results (see the later section on reliability), measure the performance of one athlete at a time (unless otherwise specified in the protocol), appear meaningful, be of suitable difficulty, be able to differentiate between various levels of ability, permit accurate scoring, include a sufficient number of trials, and withstand the test of statistical evaluation. Given the choice between two valid tests, consideration should be given to simplicity and economy of test administration.

Construct Validity

Construct validity is the ability of a test to represent the underlying construct (the theory developed to organize and explain some aspects of existing knowledge and observations). Construct validity refers to overall validity, or the extent to which the test actually measures what it was designed to measure (24). Face validity, content validity, and criterion validity, defined next, are secondary to and provide evidence for construct validity. Within construct validity, there are two important subtypes: **convergent validity** and **discriminant validity**. Both are crucial in the development and evaluation of measurement tools. Convergent validity is evidenced by high positive correlations between results of the test being assessed and those of the recognized measure of the construct (the “gold standard”). A test may be preferable to the gold standard if it exhibits convergent validity with the standard but is less demanding in terms of time, equipment, expense, or expertise. Convergent validity would be demonstrated if scores on the 1RM back squat test and the isometric squat test (requiring force plate technology) are highly correlated, indicating that both tests are effectively measuring the same underlying construct (lower body strength).

Discriminant validity is the ability of a test to distinguish between two different constructs and is evidenced by a low correlation between the results of the test and those of tests of a different construct (2). Discriminant validity would be established if scores on the 1RM back squat test do not show a correlation with $\dot{V}O_2\text{max}$ scores, indicating that the squat test is specifically measuring strength and not endurance (two different constructs).

Face Validity

Face validity is the appearance to the athlete and other casual observers that the test measures what it is purported to measure. If a test or test item has face validity, the athlete is more likely to respond to it positively (1). The assessment of face validity is

generally informal and nonquantitative. In other fields, such as psychology, tests may be deliberately constructed to have poor face validity because if examinees realize what a test or test item is supposed to measure, they can answer deceptively to manipulate their scores. For tests of basic athletic abilities, however, face validity is desirable based on the assumption that anyone taking a test of physical ability wants to do well and is thus motivated by a test that appears to measure a relevant capability.

Content Validity

Content validity is the assessment by experts that the testing covers all relevant subtopics or component abilities in appropriate proportions (1). For athletic testing, these include all the component abilities needed for a particular sport or sport position. Examples of component abilities in athletics are jumping ability, sprinting ability, and lower body strength (37). For example, a test battery for potential soccer players should include, at minimum, tests of sprinting speed, agility, aerobic endurance, and kicking power. To ensure content validity, the test developer should list the ability components to be assessed and make sure they are all represented on the test. In addition, the proportion of the total score attributable to a particular component ability should be proportional to the importance of that component to total performance. While the terms *face validity* and *content validity* are sometimes used interchangeably, the latter relates to actual validity while the former relates to the appearance of validity to nonexperts (1).

Criterion Validity

Criterion validity is the extent to which test scores are associated with some other measure of the same ability. There are two types of criterion validity: concurrent and predictive.

Concurrent validity is the extent to which test scores are associated with those of other accepted tests that measure the same ability. Criterion validity is often estimated statistically. For example,

a Pearson product–moment correlation coefficient based on the scores on a new body fat assessment device and those from dual-energy X-ray absorptiometry would provide a measure of the concurrent validity of the new test.

Predictive validity is the extent to which the test score corresponds with future behavior or performance. This can be measured through comparison of a test score with some measure of success in the sport itself. For example, one could calculate the statistical correlation between the overall score on a battery of tests used to assess potential for basketball and a measurement of actual basketball performance as indicated by a composite of such quantities as points scored, rebounds, assists, blocked shots, forced turnovers, and steals.

Reliability

Reliability is a measure of the degree of consistency or repeatability of a test (2, 8). If an athlete whose ability does not change is measured two times and achieves the same score, the test is perfectly reliable. With an unreliable test, an individual could obtain a high score on one day and a low score on another. A test must be reliable to be valid, because highly variable results have little meaning. However, even a reliable test may not be valid, because the test may not measure what it is supposed to measure. For example, both the 60-meter (66 yard) sprint and the 1.5-mile (2.4 km) run are reliable field tests, but only the 1.5-mile run is considered a valid field test for aerobic fitness. It is also possible for a test to be highly reliable for one group (e.g., college tennis players) but only moderately reliable for another group (e.g., grade school tennis players) because of differences in physical or emotional maturity and skill level, which can affect test performance.

There are several ways to determine the reliability of a test; the most obvious one is to administer the same test several times to the same group of athletes. Statistical correlation, such as the **intraclass-correlation coefficient**, of the scores from two test

administrations provides a measure of **test–retest reliability**. Any difference between the two sets of scores represents measurement error. Another statistic that can be calculated is the **typical error of measurement** (TE), which includes both the equipment error and biological variation of athletes (18). The difference between two sets of scores can arise from a number of different factors (2):

- Intrasubject (within subjects) variability
- Lack of interrater (between raters) reliability or agreement
- Intrarater (within raters) variability
- Failure of the test itself to provide consistent results

Reliability is a measure of the degree of consistency or repeatability of a test. A test must be reliable to be valid, because highly variable results have little meaning.

Intrasubject variability is a lack of consistent performance by the person being tested. **Interrater reliability**, also referred to as **objectivity** or **interrater agreement** (2), is the degree to which different raters agree in their test results over time or on repeated occasions; it is a measure of consistency. A clearly defined scoring system and competent scorers who are trained and experienced with the test are essential to enhance interrater reliability. For example, even a test that appears simple, such as timing a 40-yard (37 m) sprint with a stopwatch, can exhibit both random and systematic error if the timer is not trained and experienced. Sprint times obtained using handheld stopwatches are typically shorter than those obtained using automatic timers, because raters using stopwatches exhibit reaction-time delay when pressing the start button in response to the gun but do not delay in pressing the button at the finish line because they can see the athlete approaching. Interrater reliability is particularly important if different scorers administer tests to different subgroups of athletes. A subgroup with a relatively lenient scorer will have artificially inflated

scores. To get an accurate measure of improvement, the same scorer should test a group at the beginning and the end of the training period. If there are two scorers and the scorer at the beginning is more or less lenient than the scorer at the end, the resulting measurements may be worthless for comparative purposes. Consider a situation in which an athlete is tested in the squat. If the pretest scorer is more lenient (requiring less depth on the squat) than the posttest scorer, the athlete may achieve a lower test score on the posttest despite having made a significant improvement in strength.

Sources of interrater differences include variations in calibrating testing devices, preparing athletes, and running the test. Different testers may motivate athletes to different degrees, based on factors such as personality, status, physical appearance, demeanor, and sex. A common scenario that increases interrater variability occurs when the strength and conditioning professional tests some of the athletes while an assistant tests others. The athletes may be inspired to do better on the tests administered by the strength and conditioning professional.

The Scientific Process

Strength and conditioning professionals should establish a sound scientific process when it comes to selecting, administering, interpreting, and applying physical tests with their athletes. The scientific process of testing should begin with identifying clear questions that physical testing can answer, based on observations specific to the context of the athlete and his or her sport. Without establishing clear questions (the rationale) for physical testing, there is a risk of testing athletes for the sake of it, which is both inefficient and potentially unethical. A proposed framework for scientific process of physical test selection, with an example of a common performance question, is shown in [figure 13.1](#).

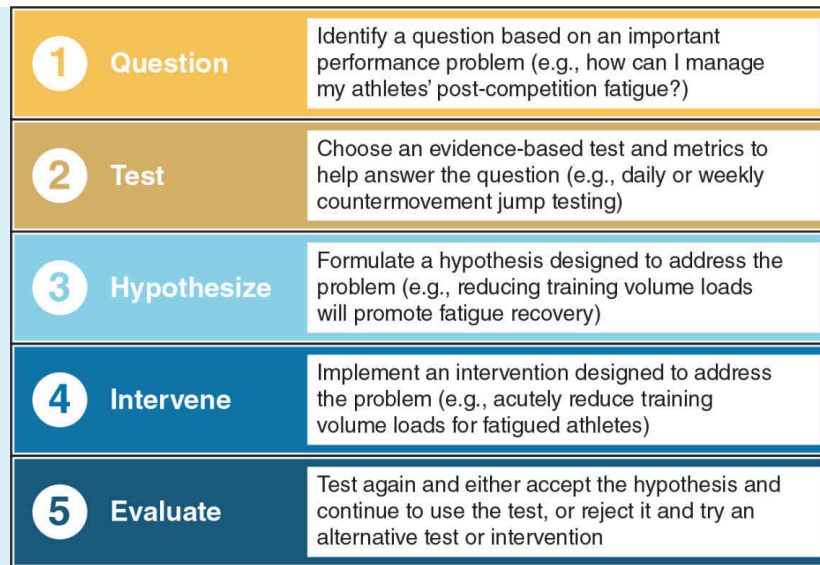


Figure 13.1 A proposed framework for the scientific process of physical test selection.

Intrarater variability is the lack of consistent scores by a given tester. This differs from interrater reliability, which refers to the degree of agreement between different testers. For example, a strength and conditioning professional eager to see improvement may unintentionally be more lenient on a posttest than on a pretest. Other causes of intrarater variability include inadequate training; inattentiveness; lack of concentration; or failure to follow standardized procedures for device calibration, athlete preparation, test administration, or test scoring. To avoid such problems, accurate and consistent athletic testing should be a priority for all strength and conditioning professionals.

Finally, sometimes the test itself might fail to provide consistent results. This may occur if a physical performance test requires a technique in which the athlete has not developed consistency. More technique-intensive tests generally exhibit greater variability in results and require more pretest practice to produce consistency.

Test Selection

When evaluating tests for high levels of validity and reliability, the strength and conditioning professional must rely on her or his knowledge and practical experience with the sport. The strength and conditioning professional must consider sport specificity (e.g., metabolic energy systems, biomechanical movement patterns), athlete experience, training status, age, and environmental factors when selecting tests.

When considering test selection, it is imperative to classify the exercises to be tested. The categorization of exercises facilitates the creation of a comprehensive testing battery that accurately evaluates an athlete's capabilities across various movement patterns and muscle groups. Understanding the classification system assists strength and conditioning professionals in designing targeted training programs tailored to individual needs and goals. The classification of exercises for assessment typically encompasses several key dimensions of specificity:

- *Movement patterns.* Exercises can be classified based on the primary movement patterns they involve, such as pushing, pulling, squatting, hinging, rotating, and stabilizing (7, 8). Each movement pattern places specific demands on the musculoskeletal system and requires distinct biomechanical considerations. By including tests that cover a spectrum of movement patterns, practitioners can obtain a complete understanding of an athlete's functional capabilities.
- *Muscle groups targeted.* Another crucial dimension for classification is the muscle groups targeted during an exercise. This classification helps identify potential muscle imbalances or weaknesses that may predispose athletes to injury or limit their performance.
- *Primary energy system.* Exercises can also be categorized based on the primary energy system they predominantly

engage, such as the phosphagen system, glycolytic system, or oxidative system. Different energy systems contribute to various aspects of athletic performance, including power, strength, and endurance. Assessing performance across these energy systems provides valuable insights into an athlete's metabolic capacity and helps tailor conditioning strategies accordingly.

- *Skill level and complexity.* Exercise classification should also consider the skill level and complexity involved in performing a particular movement. Basic compound movements like squats, deadlifts, and presses may serve as foundational tests for assessing general strength and movement proficiency. Conversely, more advanced or sport-specific movements require greater neuromuscular coordination and may be included to evaluate athletic prowess in specific contexts.

By systematically categorizing exercises based on these dimensions, strength and conditioning professionals can construct a targeted testing battery that effectively evaluates an athlete's strength, power, endurance, and movement quality. This comprehensive approach not only identifies areas of improvement but also guides the development of evidence-based training programs aimed at optimizing athletic performance and reducing the risk of injury. [Table 13.1](#) details common tests used to assess physical characteristics of various sports.

Metabolic Energy System Specificity

A valid test must emulate the energy requirements of the sport for which ability is being assessed. Thus, the strength and conditioning professional should have a thorough understanding of the basic energy systems (phosphagen, glycolytic, and oxidative) and their interrelationships in order to apply the principle of specificity when choosing or designing valid tests to measure athletic ability for specific sports (10, 13, 19, 36). For example, in choosing an appropriate test for running ability in basketball, the strength and

conditioning professional must understand that basketball is predominantly an anaerobic running sport (4, 26) and also be familiar with the distances and directions of sprints in a basketball game. It is best for the tests to simulate the physical movements and energy demands of a real game.

TABLE 13.1 Common Tests Used to Assess Physical Characteristics of Various Sports

Physical characteristic	Test name	Sport application examples	Description
Aerobic capacity	Yo-Yo intermittent recovery test	Soccer, hockey	Measures aerobic fitness and ability to recover between bouts of high-intensity exercise
	$\dot{V}O_2$ max test	Aerobic endurance sports	Measures maximum oxygen uptake during incremental exercise, indicating aerobic endurance capacity
Agility	T-test	Soccer, basketball, American football	Assesses agility by measuring the time taken to navigate a T-shaped course
	Illinois test	Soccer, rugby, American football	Measures agility and speed by timing the performance through a set course with changes of direction
Anaerobic capacity	Wingate anaerobic test	Cycling, hockey	Assesses anaerobic power and capacity by measuring power output on a cycle ergometer over 30 s
Anthropometry	Height, body mass, arm span, circumferences, and limb lengths	General	Measures the dimensions and proportions of the human body
Balance and stability	Balance error scoring system (BESS)	General	Assesses static balance by counting errors during six 20-s stance conditions performed on firm and foam surfaces, with

Physical characteristic	Test name	Sport application examples	Description
			errors including opening eyes, lifting hands from hips, stepping or stumbling, or moving from the test position
	Star excursion balance test (SEBT)	General	Evaluates dynamic balance and neuromuscular control by having individuals maintain single-leg stance while reaching with their free leg in eight different directions arranged in a star pattern, measuring their maximum reach distance in each direction
Body composition	Air displacement plethysmography	General	Measures body composition by calculating body volume and density through the displacement of air in a sealed chamber (e.g., the Bod Pod device)
	Bioelectrical impedance	General	Estimates body fat by measuring the resistance to a small electrical current passed through the body
	Dual X-ray absorptiometry (DXA)	General	Measures body composition by using low-dose X-ray beams at two different energy levels to distinguish between bone, fat tissue, and lean mass throughout the body
	Skinfolds	General	Estimates body fat percentage by using calipers to measure subcutaneous fat thickness at specific sites on the body, which are then entered into prediction equations
	Underwater weighing	General	Measures body composition by submerging a person in water and using their underwater

Physical characteristic	Test name	Sport application examples	Description
			weight to calculate body density and estimate body fat percentage based on Archimedes' principle
Flexibility	Sit and reach	General	Assesses hamstring and lower back flexibility by measuring how far a person can reach forward while sitting with legs extended
	Overhead squat	General	Used as a part of movement screens to assess bilateral mobility of hips, knees, and ankles along with shoulder and thoracic spine
Muscular endurance	Plank	General	Measures core strength and endurance by maintaining a plank position for as long as possible
	Push-up	General	Measures upper body muscular endurance by completing as many repetitions as possible in a certain time or to the point of exhaustion
Maximum muscular power	Vertical jump	Basketball, volleyball, American football	Measures lower body explosive power by the height achieved in a vertical jump
	Standing long jump	Track and field, American football	Measures horizontal explosive power by the distance jumped from a standing position
	1RM power clean	Weightlifting, track and field, rugby, American football	Assesses the maximum weight that can be lifted during the power clean exercise
Maximum muscular	1RM for the upper body (e.g., bench	American football, rugby,	Measures maximal upper body strength by determining the

Physical characteristic	Test name	Sport application examples	Description
strength	press or bench pull)	powerlifting	maximum weight that can be lifted once
	1RM for the lower body (e.g., back squat, front squat, deadlift)	American football, rugby, powerlifting	Measures maximal lower body strength by determining the maximum weight that can be lifted once
	Isometric mid-thigh pull	Weightlifting, track and field, American football, rugby	Assesses maximal isometric strength by pulling on an immovable bar
Speed	40-yard (37 m) sprint	American football, rugby	Measures sprinting speed over a specific distance

Biomechanical Movement Pattern Specificity

All else being equal, the more similar the test is to an important movement in the sport, the better. Sports differ in their physical demands. For example, the vertical jump test is very specific to basketball and volleyball, both of which involve vertical jumping during play, but less relevant to hockey, which does not involve vertical jumping. Positions within a sport differ as well. An American football defensive lineman needs pushing strength to move opposing linemen out of the way and 5- to 15-yard (5 to 14 m) sprint speed to reach the opposing quarterback, while a wide receiver depends less on pushing strength but must be able to sprint 30 to 100 yards (27-91 m) quickly. Thus, the bench press and 10-yard (9 m) sprint test would be more relevant to the lineman, while sprint tests of 30 to 100 yards (27-91 m) would be more relevant to the wide receiver.

For a test to be valid, it must emulate the energy requirements and important movements of the sport for which ability is being tested.

Experience and Training Status

For a well-trained, experienced athlete, a technique-intensive test may be appropriate because it can be very sport specific, and one can assume that poor technique will not impair performance of the test. However, this assumption cannot be made for an athlete just learning or trying out for a sport. The number of single-leg hops needed to travel 27 yards (25 m) may represent a valid and reliable test of plyometric strength for an experienced long jumper but not for a novice (11).

Testers must also consider the training status of the athletes being tested. It would not be ideal, for example, to ask a baseball player to perform a 3-mile (4.8 km) run test a week before the beginning of fall practice, because the player has probably been doing interval training and relatively short runs (32). A lower body strength test using the parallel squat would not be an ideal test for an athlete who has trained using the leg press exclusively.

Age and Sex

Both age and sex can affect the validity and reliability of a test. For example, the 1.5-mile (2.4 km) run may be a valid and reliable field test of aerobic power for college-aged men and women (21) but may not be appropriate for preadolescents because of their probable lack of experience and interest in sustained running (30). A test of the maximum number of chin-ups that can be performed may be a valid test of elbow flexion muscular endurance for male wrestlers, but it may not be as valid for females due to differences in upper body pulling strength (27). The test may not be capable of differentiating muscular endurance levels in females. Therefore, using a modified prone pull-up with the feet supported may be a more valid test.

Environmental Factors

It is necessary to consider the environment when selecting and administering tests of basic athletic ability. High ambient

temperature, especially in combination with high humidity, can impair endurance exercise performance, pose health risks, and lower the validity of an aerobic endurance exercise test. Aerobic endurance performance (29, 35) and intermittent sprint performance (17) may be impaired when the temperature approaches 80 °F (27 °C), especially if the humidity exceeds 50% (20). The effects of temperature and humidity on testing performance can create problems for comparing the results of tests administered at different times of year, on different days, and even at different times of day. For example, the maximal oxygen uptake of an athlete impaired by the heat is underestimated by the 1.5-mile (2.4 km) run test. Run times can also be impaired by cold temperatures. Thus, outdoor aerobic endurance tests may be inappropriate at locations characterized by wide fluctuations in temperature. In such places, aerobic endurance tests can be administered on an indoor track, if available, or with a treadmill or stationary cycle.

Altitude can also impair performance on aerobic endurance tests, although not on tests of strength and power (24). Norms on aerobic endurance tests should be adjusted when testing at altitudes exceeding 1,900 feet (579 m). Up to about 9,000 feet (2,743 m), maximal oxygen uptake declines by approximately 5% for each 3,000 feet (914 m) of elevation (25). At even higher altitudes, maximal oxygen uptake declines more sharply. Athletes who arrive at a relatively high altitude after living near sea level for an extended period of time should be given at least 10 days to acclimate before undergoing aerobic endurance tests (34). For all testing, it is good practice to measure and document the environmental conditions and then to consider these factors when interpreting the results.

Athletes' experience, training status, age, and sex can affect test performance, so these factors should be considered in test selection. Environmental factors such as temperature, humidity, and altitude can also influence test performance, so testers should try to

standardize environmental conditions as much as possible.

Test Administration

To achieve accurate test results, tests must be administered safely, correctly, and in an organized manner. Strength and conditioning professionals should ensure the health and safety of athletes; testers should be carefully selected and trained; tests should be well organized and administered efficiently; and athletes should be properly prepared and instructed.

Health and Safety Considerations

Even though all athletes should be medically cleared before being permitted to physically train and compete, the strength and conditioning professional must be aware of testing conditions that can threaten the health of athletes and be observant of signs and symptoms of health problems that warrant exclusion from testing (5). The strength and conditioning professional must remain attentive to the health status of athletes, especially before, during, and after maximal exertions that occur during training, testing, and competition. Strenuous exercise, such as maximal runs or 1-repetition maximum (1RM) tests, can uncover or worsen existing heart problems, such as impaired blood flow to the heart muscle and irregular heartbeats. Standard medical screening cannot always reveal hidden heart problems, which occasionally result in fatality among young athletes (30). Heat injury is also a risk during heavy physical exertion in hot environments, especially when humidity is high. Athletes should wear light clothing in warm weather and drink water ad libitum according to the dictates of thirst before and during heavy physical exertion in the heat (refer to [chapters 10](#) and [11](#) for more detailed hydration guidelines). Musculoskeletal injuries can also be a problem. If symptoms are ignored, recovery can be greatly delayed.

Medical referral may be warranted for an athlete who persistently has any of the following symptoms: chest pressure, pain, or discomfort; listlessness; light-headedness; dizziness; confusion;

headache; deeply reddened or cold and clammy skin; irregular pulse; bone or joint pain; blurred vision; nausea; or shortness of breath, rapid pulse, or weakness either not commensurate with the level of exertion or unresponsive to rest. Such symptoms can occur long after exercise is terminated. Even symptoms that occur only once, if severe (such as loss of consciousness), call for immediate medical attention.

When aerobic endurance exercise tests are being administered in a hot environment, caution must be observed to protect both the health and safety of the athlete and the validity of the test. [Figure 13.2](#) lists temperature limits at various ranges of relative humidity for strenuous exercise testing, and the sidebar lists guidelines for aerobic endurance testing in the heat.

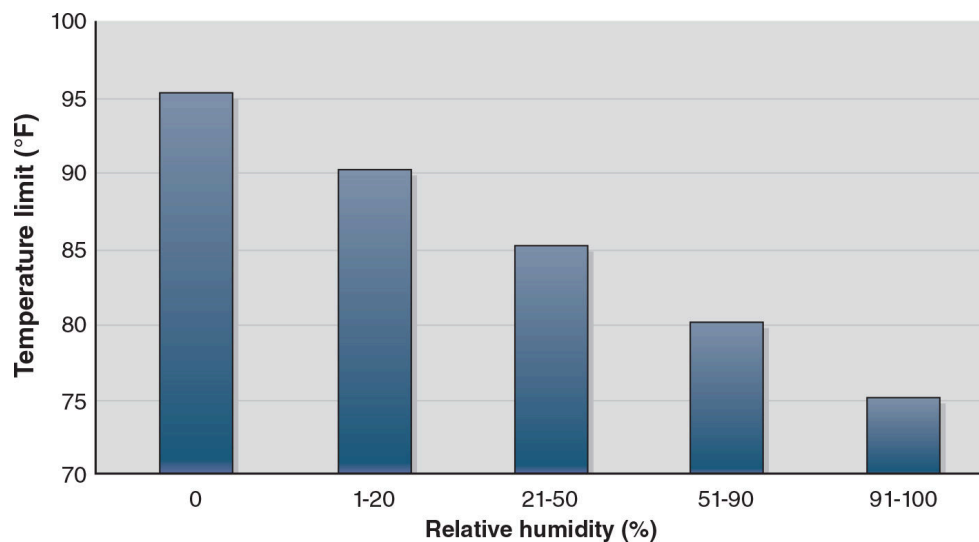


FIGURE 13.2 Temperature limits at various ranges of relative humidity for strenuous exercise testing.

Data from McArdle et al. (2023).

Aerobic Endurance Testing in the Heat

Follow these guidelines to minimize health risks and obtain accurate results when testing athletes in hot environments (9, 23, 30):

- During the weeks before the test, athletes should engage in enough training to establish a baseline of fitness in the activity being tested. Strength and conditioning professionals should be aware that responses to exercise in the heat are highly individual.
- Avoid testing under extreme combinations of heat and humidity. [Figure 13.2](#) illustrates the combinations of temperature and humidity at which heat injury risk is present.
- Using temperature limits of at least 5 °F (3 °C) below those listed is recommended, especially on sunny days, to provide a safety margin and enable better test performance. On days when the temperature has exceeded or is expected to exceed the recommended limits, indoor facilities should be used if available, or testing should be conducted during morning or early evening hours when temperatures are acceptable.
- The athletes, especially those coming from cool climates, should be acclimated to the heat and humidity for at least one week before testing. Start with short workouts and progress to workouts of longer duration.
- Athletes should make sure they are well hydrated in the 24-hour period preceding aerobic endurance testing in the heat. A good indication of adequate hydration is a plentiful volume of clear urine (refer to [chapter 10](#) for more detailed preexercise hydration guidelines). Salt tablets should generally be avoided.
- Athletes should be encouraged to drink during exercise in the heat. Plain water is most appropriate for exercise up to 1 hour in duration. Snack breaks during longer durations of training can help replenish fluids and be important in replacing electrolytes. The amount and rate of fluid replacement depend on individual sweat rates, duration of exercise, and environment. Refer to [chapter 10](#) for specific guidelines and how to calculate sweat rate if needed.
- Athletes should wear a light-colored, loose-fitting tank top and shorts that are breathable (i.e., synthetic porous, moisture-wicking material). Male athletes may be allowed to go shirtless.
- Heart rate may be monitored to detect reactions to the heat.
- Be attentive to possible symptoms of heatstroke or heat exhaustion: cramps, nausea, dizziness, difficulty in walking or standing, faintness, garbled speech, lack of sweat, red or ashen skin, and goose bumps.
- Be aware of the symptoms of hyponatremia or water intoxication, a potentially fatal condition in which excess water intake reduces blood sodium to dangerously low levels. Symptoms may include extremely diluted urine in combination with bloated skin, altered consciousness, or loss of consciousness, with no increase in body temperature. A victim of hyponatremia should never be given fluid and should be treated by a physician.
- Proficient medical coverage should be readily available so that an athlete encountering a test-related health problem can be very rapidly treated or

evacuated (or both).

Selection and Training of Testers

Test administrators should be well trained and should have a thorough understanding of all testing procedures and protocols. The testing supervisor should make sure that all novice personnel perform and score all tests correctly, as in timing sprint speed with a stopwatch or determining a 1RM back squat. It is essential that all testers have sufficient practice so that the scores they obtain correlate closely with those produced by experienced and reliable personnel. The testers should be trained to explain and administer the tests as consistently as possible. Test reliability is impaired, for example, if one test administrator provides considerable verbal encouragement to a group of athletes while another tester provides no verbal encouragement to another group. Administrators should have a checklist of materials needed for testing and written test protocols to refer to if questions arise during the testing process.

Recording Forms

Scoring forms, whether paper based or digital, should be developed before the testing session and should have space for all test results and comments. Factors such as environmental conditions should be documented. The specific details of setup for the testing should also be noted. For example, for a 1RM squat, the tester can note the pin height that was used for the barbell. This allows test time to be used more efficiently and reduces the incidence of recording errors.

Test Format

A well-organized testing session, in which the athletes are aware of the purpose and procedures of the testing, will enhance the reliability of test measures. Reliable measures obtained from valid tests are a great asset in assessing fitness levels and evaluating changes over a period of time.

Test planning must address such issues as whether athletes will be tested all at once or in groups and whether the same person will administer a given test to all athletes. The main consideration here is the number of athletes that are being tested. It is preferable to have the same person administer a given test if time and schedules permit, because this eliminates the issue of interrater reliability. If this is not feasible, the test supervisor can allow simple, well-defined tests (e.g., counting correct push-ups) to be administered by different testers and tests requiring schooled judgment (as of proper form in the squat) to be scored by the most skilled and experienced personnel. As a rule, each tester should administer only one test at a time, especially when the tests require complex judgments. It is permissible to have one tester alternate between two testing stations to avoid wasting time as athletes get ready. However, the tester must focus on only one test at a time. Planning and practicing the testing sessions beforehand will go a long way toward ensuring an efficiently run testing session.

Testing Batteries and Multiple Testing Trials

When time is limited and the group of athletes is large, duplicate test setups may be employed to make efficient use of testing time. For example, when one is conducting the 300-yard (274 m) shuttle, two test courses can be set up (16). A tester can administer up to two nonfatiguing tests in sequence to an athlete as long as test reliability can be maintained. For example, at a two-test lower body power station staffed by only one tester, the athlete can perform the vertical jump test and the static jump test immediately afterward.

When multiple trials of a test (e.g., the repeated trials it takes to find a 1RM) or a battery of tests is performed, allow complete recovery between trials (31). There should be at least 2 minutes of rest between attempts that are not close to the athlete's maximum and 3 minutes between attempts that are close to the maximum, as judged by the relative difficulty of the previous trial or testing set

(21). When administering a **test battery** (e.g., one in which wrestlers perform maximal repetition pull-up and push-up tests for assessment of local muscular endurance), tests should be separated by at least 5 minutes to prevent the effects of fatigue from confounding test results (also see the following section, Sequence of Tests).

Sequence of Tests

Knowledge of exercise science can help determine the proper order of tests and the duration of rest periods between tests to ensure test reliability. The fundamental principle with test sequencing should be that one test should not affect the performance of a subsequent test. This should allow for optimal performance in each test and also allows for valid comparisons with previous testing results. For example, a test that maximally taxes the phosphagen energy system requires 3 to 5 minutes of rest for complete recovery (6, 12), whereas a maximal test of the anaerobic glycolytic energy system requires at least 1 hour for complete recovery (10). Therefore, tests requiring high-skill movements, such as agility tests, should be administered before tests that are likely to produce fatigue and confound the results of subsequent tests. A logical sequence, although there are some variations, is to administer tests in this order:

1. Nonfatiguing tests (e.g., height, weight, flexibility, skinfolds and girth measurements, vertical jump)
2. Agility tests (e.g., T-test, 5-10-5 test, 5-0-5 test, Illinois test)
3. Maximum power and strength tests (e.g., 1RM power clean, 1RM squat, 1RM bench press)
4. Sprint tests (e.g., 40 m sprint with split times at 10 m and 20 m)
5. Local muscular endurance tests (e.g., push-up test, partial curl-up, YMCA bench press test)

6. Fatiguing anaerobic capacity tests (e.g., 300-yard [274 m] shuttle, 90 s box hop test)
7. Aerobic capacity tests (e.g., 1.5-mile [2.4 km] run, 12-minute run test, Yo-Yo intermittent recovery test, maximal aerobic speed test, and 30-15 intermittent fitness test)

The test order should also be designed to require minimal recovery time between tests, allowing for a more efficient testing session. An effort should be made to administer fatiguing anaerobic capacity tests and aerobic tests on a different day than the other tests. However, if performed on the same day, these tests should be performed last, after an extended rest period.

It is important that test sessions be conducted at the same time of day to avoid fluctuations in physiological responses due to differences in circadian rhythm (33). It is also recommended that tests be conducted indoors, which will allow for climate and testing surfaces to remain more consistent.

The order of tests should be designed in such a way that the completion of one test does not adversely affect performance in subsequent tests.

Preparing Athletes for Testing

The date, time, and purpose of a test battery should be announced in advance to allow athletes to prepare physically and mentally. To maximize test reliability, athletes should be familiar with test content and procedures. Athletes should also be in a state of **readiness** to perform. Athlete readiness has been defined as “the state of being fully prepared to train and perform physically and mentally” (3). A short, supervised pretest practice or familiarization session one to three days before the test, in which the athletes exert themselves at somewhat less than full intensity, is often beneficial for athletes who either have never participated in the test before or have not done so for a long time. One strategy to promote test familiarization among

athletes could be to incorporate familiarization of the tests into preceding training sessions, even if only as part of their dynamic warm-up.

The instructions should cover the purpose of the test, how it is to be performed, the amount of warm-up recommended, the number of practice attempts allowed, the number of trials, test scoring, criteria for disallowing attempts, and recommendations for maximizing performance. The instructions that are given to the athletes need to be clear and concise because this will increase the reliability and objectivity of the test (25).

As an important supplement to reading test instructions aloud, the test administrator or a competent assistant should demonstrate proper test performance when possible. The athletes should be given opportunities to ask questions before and after the demonstration. The test administrator should anticipate questions and have answers prepared. It is important to motivate all athletes equally rather than giving special encouragement to only some. Whenever possible, tell athletes their test scores immediately after each trial to motivate them to perform better on subsequent trials (2).

Reliability of testing improves with pretest warm-up (2). An appropriately organized warm-up consists of a general warm-up followed by a specific warm-up. Both types of warm-ups include body movements similar to those involved in the test. An organized, instructor-led general warm-up ensures uniformity. It is acceptable to allow two or three activity-specific warm-up trials, depending on the test protocol, and have subsequent trials actually count toward the score. Depending on the test protocol, the score can be the best or the average of the post-warm-up trials (2).

General and specific warm-ups performed before a test can increase the test's reliability.

Administer a supervised cool-down period to athletes following tests that dramatically increase heart rate and at the completion of the test battery. For example, after the 300-yard (274 m) shuttle, the athlete should not sit or lie down; active recovery using low-intensity movement and light stretching will enhance the recovery process (10).

Conclusion

Tests and measurements can be used to assess athletic talent, identify physical capacities in need of improvement, provide reference values to evaluate the effectiveness of a training program, and set realistic training goals. To optimize test quality, testers must understand and consider validity and reliability. Test selection involves consideration of the physiological energy systems required by the sport; movement specificity; and the athletes' experience, training status, age, and sex. Testers must also consider environmental factors such as temperature, humidity, and altitude before administering tests. Strength and conditioning professionals must always remain conscious of potential health risks during testing and attentive to signs and symptoms of possible health problems that require medical referral. Testers must be carefully selected and well trained, and the testing session must be well planned and organized using the appropriate testing sequence. Consistent and effective preparation of athletes for testing is essential.

KEY TERMS

- concurrent validity
- construct validity
- content validity
- convergent validity
- criterion validity
- discriminant validity
- evaluation
- face validity
- field test
- formative evaluation
- interrater agreement
- interrater reliability
- intraclass-correlation coefficient

intrarater variability
intrasubject variability
measurement
midtest
monitoring
objectivity
posttest
predictive validity
pretest
readiness
reliability
talent identification
test
test battery
test–retest reliability
typical error of measurement
validity

STUDY QUESTIONS

1. Regular assessment through evidence-based testing is important for monitoring progress in athletes because of which of the following reasons?
 - a. It ensures the athlete maintains a consistent training routine.
 - b. It provides subjective feedback from the athlete about their feelings.
 - c. It allows coaches to track changes and make data-driven program modification.
 - d. It guarantees immediate improvement in athletic performance.
2. Reliability is crucial in test selection because it ensures that the test does which of the following?
 - a. creates a challenge for the athlete

- b. produces consistent results over multiple trials
 - c. provides a way to quickly administer the test
 - d. reduces the expense of the test
3. Which of the following tests is most appropriate for assessing aerobic endurance in a soccer player?
- a. 1RM bench press
 - b. 40-yard sprint
 - c. $\dot{V}O_2$ max test
 - d. vertical jump
4. When assessing the validity of a test, which aspect is the most critical?
- a. ability to be repeated consistently
 - b. alignment with the sport-specific demands
 - c. affordability and ease of implementation
 - d. capacity to accommodate large groups
5. To evaluate an athlete's lower body power, which of the following tests would most accurately provide valid results for this purpose?
- a. T-test
 - b. vertical jump
 - c. 40-yard sprint
 - d. 1RM deadlift